Infinite Sequences and Series



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11.7 Strategy for Testing Series



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Strategy for Testing Series

- **1.** If the series is of the form $\Sigma\ 1/n^p$, it is a *p*-series, which we know to be convergent if $p > 1$ and divergent if $p \leq 1$.

- **2.** If the series has the form $\Sigma\ ar^{n-1}$ or $\Sigma\ ar^n$, it is a geometric series, which converges if $|r| < 1$ and diverges if $|r| \geq 1$. Some preliminary algebraic manipulation may be required to bring the series into this form.

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Strategy for Testing Series

- **3.** If the series has a form that is similar to a *p*-series or a geometric series, then one of the comparison tests should be considered. In particular, if a_n is a rational function or an algebraic function of n (involving roots of polynomials), then the series should be compared with a *p*-series.

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The comparison tests apply only to series with positive terms, but if Σa_n has some negative terms, then we terms, then we can apply the Comparison Test to $\Sigma \mid a_n \mid$ and test for absolute convergence.

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Strategy for Testing Series

- **4.** If you can see at a glance that $\lim_{n\to\infty} a_n \neq 0$ then the Test for Divergence should be used.

- **5.** If the series is of the form $\Sigma(-1)^{n-1}b_n$ or $\Sigma(-1)^n b_n$, then the Alternating Series Test is an obvious possibility.

- **6.** Series that involve factorials or other products (including a constant raised to the *n*th power) are often conveniently tested using the Ratio Test. Bear in mind that $|a_{n+1}/a_n| \to 1$ as $n \to \infty$ for all *p*-series and therefore all rational or algebraic functions of *n*. Thus the Ratio Test should not be used for such series.

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Strategy for Testing Series

- **7.** If a_n is of the form $(b_n)^n$, then the Root Test may be useful.

- **8.** If $a_n = f(n)$, where $\int_1^\infty f(x)\,dx$ is easily evaluated, then the Integral Test is effective (assuming the hypotheses of this test are satisfied).

- In the following examples we don't work out all the details but simply indicate which tests should be used.

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Example 1

$$\sum_{n=1}^{\infty} \frac{n-1}{2n+1}$$

- Since as $a_n \to \frac{1}{2} \neq 0$ as $n \to \infty$, we should use the Test for Divergence.

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Example

$$\sum_{n=1}^{\infty} \frac{\sqrt{n^3 + 1}}{3n^3 + 4n^2 + 2}$$

- Since a_n is an algebraic function of n, we compare the given series with a p-series.

$\sum b_n,$

- The comparisc mit Comparison Te
- where

$$b_n = \frac{\sqrt{n^3}}{3n^3} = \frac{n^{3/2}}{3n^3} = \frac{1}{3n^{3/2}}$$

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Example

$$\sum_{n=1}^{\infty} (-1)^n \frac{n^3}{n^4 + 1}$$

- Since the series is alternating, we use the Alternating Series Test.

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Example

$$\sum_{k=1}^{\infty} \frac{2^k}{k!}$$

- Since the series involves $k!$, we use the Ratio Test.

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Example

$$\sum_{n=1}^{\infty} \frac{1}{2 + 3^n}$$

- Since the series is closely related to the geometric series
- $\sum 1/3^n$, we use the Comparison Test.

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